MTH2008 Real Analysis
January 2021 Exam Solutions

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SECTION A

1. (a) Consider the following subset of the real numbers

$$S = (-1, 2) \cup [3, \infty).$$

i. Is the set S open? Is the set S closed? In each case justify your answer.

This set is neither open nor closed.
It is not open since 3 is not an interior point.
It is not closed since it does not include all of its limit points: -1 and 2 are limit points that are not in the set. Alternatively, the complement is

$$S^c = (-\infty, -1] \cup [2, 3)$$

which is not open, since -1 and 2 are not interior points.

ii. Determine the interior of S and the set of boundary points of S.

$$S^0 = (-1, 2) \cup (3, \infty).$$
$$\partial S = \{-1, 2, 3\}.$$

(b) Prove, using an "ϵ-δ" argument, that the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \dfrac{1}{x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

is not continuous at $x_0 = 0$.

For continuity at 0 we would require that for all $\epsilon > 0$, there exists a $\delta > 0$ such that for all $0 < |x - 0| < \delta$,

$$\left| \frac{1}{x^2} - 0 \right| < \epsilon.$$

However, assuming such a δ exists, we can consider a point x_1 such that $0 < |x_1 - 0| < \delta$ and such that $0 < |x_1| < \frac{1}{\sqrt{2\epsilon}}$, but then we have

$$\left| \frac{1}{x_1^2} \right| > 2\epsilon > \epsilon,$$

which is a contradiction.

Alternatively, we can show that $\lim_{x\to 0} f(x) = \infty$ and then since the limit does not exist f is not continuous at 0.
To that end: for any $M \in \mathbb{R}$ let $\delta = \frac{1}{\sqrt{|M|}}$, then for all x satisfying $0 < |x| < \delta$ we have

$$f(x) = \frac{1}{x^2} > \frac{1}{1/\sqrt{|M|}^2} = |M| \geq M$$

hence $\lim_{x\to 0} f(x) = \infty$.

(c) Let $\{s_n\}$ be the sequence

$$s_n = \sin\left(\frac{n\pi}{2}\right) + \cos(n\pi).$$

Find all numbers L in the extended reals that are limits of some subsequence of $\{s_n\}$, and, for each such L choose a subsequence $\{s_{n_k}\}$ such that $\lim_{k\to\infty} s_{n_k} = L$. Does the sequence $\{s_n\}$ converge? Justify your answer.

We have $s_n \in \{1, 0, -2\}$ and in particular

$$s_k = \begin{cases} 1 & k = 2n \\ 0 & k = 4n+1 \\ -2 & k = 4n+3 \end{cases}$$

and so clearly

$$\lim_{n\to\infty} s_{2n} = 1$$
$$\lim_{n\to\infty} s_{4n+1} = 0$$
$$\lim_{n\to\infty} s_{4n+3} = -2.$$

If $\{s_n\}$ were convergent then every subsequence must have the same limit. Since we have subsequences with different limits, $\{s_n\}$ cannot converge.

(d) Let $\{f_n\}$ be the sequence of functions $f_n : [0,1] \to \mathbb{R}$ defined by

$$f_n(x) = \frac{x^n}{n}$$

and let $f(x) = \lim_{n\to\infty} f_n(x)$, where the limit exists. Find f. Does $\{f_n\}$ converge uniformly to f? Justify your answer.

For all $x \in [0,1]$ we have

$$|f_n(x)| = \left|\frac{x^n}{n}\right| \leq \frac{1}{n}$$

and hence

$$f(x) = \lim_{n\to\infty} f_n(x) = 0$$

and f is defined on $[0,1]$.

The convergence is uniform.
Proof: for all $\epsilon > 0$,

$$\|f_n - f\|_{[0,1]} = \sup_{x\in[0,1]} \left|\frac{x^n}{n} - 0\right| = \left|\frac{1}{n}\right| < \left|\frac{1}{N}\right| < \epsilon$$

for all $n > N$ with $N \geq \dfrac{1}{\epsilon}$ (that is, $\lim_{n\to\infty} \|f_n - f\|_{[0,1]} = 0$).

(e) Show that the function $f : \mathbb{R}^3 \to \mathbb{R}$ given by

$$f(x) = x_1^3 + x_1 x_3 + x_2^2 - 2x_2$$

for all $x = (x_1, x_2, x_3)^\intercal \in \mathbb{R}^3$ is differentiable and find the derivative. Hence determine the stationary points of f.

The Jacobian is

$$J(x) = \begin{pmatrix} D_1 f(x) & D_2 f(x) & D_3 f(x) \end{pmatrix} = \begin{pmatrix} 3x_1^2 + x_3 & 2x_2 - 2 & x_1 \end{pmatrix}$$

and since each partial derivative is continuous (as they are polynomials), this is indeed the derivative and f is differentiable.
For a stationary point we must have

$$\begin{aligned} 3x_1^2 + x_3 &= 0 \\ 2x_2 - 2 &= 0 \\ x_1 &= 0 \end{aligned}$$

which implies that $x = (0,1,0)^\intercal$ is the only stationary point.

SECTION B

2. (a) Prove or disprove (using a counterexample) the following statements.

   i. A limit point of a set S is either an interior point of S or a boundary point of S.

   True. If x_0 is a limit point then every neighbourhood U of x_0 intersects S. If every such U also intersects points in S^c, then x_0 is a boundary point. If this is not the case, then there exists a U such that $U \subset S$, and hence x_0 is an interior point.

   ii. A set S has no limit points if and only if each of its points is isolated.

   False. For example the set $S = \left\{ \frac{1}{n} \middle| n \in \mathbb{N} \right\}$ consists of isolated points, which can be seen by considering the open neighbourhoods

   $$\left(\frac{1}{n + \frac{1}{2}}, \frac{1}{n - \frac{1}{2}} \right),$$

   however, 0 is a limit point of S.

   iii. The boundary ∂S of a set S is closed.

   True. For all $x_0 \in (\partial S)^c$ there exists an open neighbourhood U of x_0 such $U \cap \partial S = \emptyset$, since otherwise if there were $x_1 \in U \cap \partial S$, then U is also a neighbourhood of $x_1 \in \partial S$ and we must have $U \cap S \neq \emptyset$ and $U \cap S^c \neq \emptyset$ which would imply that x_0 is also a boundary point. Since $U \subset (\partial S)^c$, this says that x_0 is an interior point of $(\partial S)^c$, hence $(\partial S)^c$ is open and ∂S is closed.

   (b) Suppose that $f : [a, b] \to \mathbb{R}$ is continuous and that $f(x)$ is irrational for all $x \in [a, b]$. Prove that f must be a constant function.

   Assume to the contrary that f is not constant, then there exists $x_1 < x_2$ such that $f(x_1) < f(x_2)$ (or $f(x_1) > f(x_2)$). By the density of the rationals, there exists a rational μ such that $f(x_1) < \mu < f(x_2)$ (or $f(x_1) > \mu > f(x_2)$) and by the Intermediate Value Theorem there exists a $c \in (x_1, x_2)$ such that $f(c) = \mu$. This is a contradiction to $f(x)$ being irrational for all x, hence f must be constant.

3. (a) Assume that $f'(x) > g'(x)$ for all $x \in \mathbb{R}$ and that $f(x_0) = g(x_0)$ for some $x_0 \in \mathbb{R}$.

   i. Show that $f(x) > g(x)$ for $x > x_0$ and $f(x) < g(x)$ for $x < x_0$. (*Hint: consider the function $f - g$.*)

   Since $(f-g)'(x) = f'(x) - g'(x) > 0$ for all $x \in \mathbb{R}$, using Theorem 2.76 from the notes (which follows from the Mean Value Theorem) we have that $f - g$ is an increasing function. Then since $(f-g)(x_0) = 0$ we have for $x < x_0$ that $(f-g)(x) < 0$, that is $f(x) < g(x)$, whilst for $x > x_0$ we have $(f-g)(x) > 0$, that is $f(x) > g(x)$.

   ii. Show, by providing an example for each case, that if we omit the assumption that $f(x_0) = g(x_0)$ for some $x_0 \in \mathbb{R}$, then it is possible that $f(x) > g(x)$ for all $x \in \mathbb{R}$ or that $f(x) < g(x)$ for all $x \in \mathbb{R}$.

   Consider for example $f(x) = e^x$ and $g(x) = 0$. Then $f'(x) = e^x > 0 = g'(x)$ but $f(x) > g(x)$ for all $x \in \mathbb{R}$. For the other direction of the inequality, we could take $f(x) = -e^{-x}$ and $g(x) = 0$, then $f'(x) = e^{-x} > 0 = g'(x)$, but $g(x) > f(x)$ for all $x \in \mathbb{R}$.

(b) Prove that if f is a continuous non-negative function on $[a, b]$ and $\int_a^b f(x)\,dx = 0$, then $f(x) = 0$ for all $x \in [a, b]$. (*Hint: try a proof by contradiction.*) Show by an example that the conclusion doesn't hold if we omit the assumption that f is continuous.

Assume for a contradiction that $f(x) \neq 0$ for all x, i.e. there exists an x_0 such that $f(x_0) > 0$. Since f is continuous at x_0, taking $\epsilon = f(x_0)/2 > 0$ there exists a $\delta > 0$ such that for all $|x - x_0| < \delta$ we have $|f(x) - f(x_0)| > \epsilon$, i.e. $f(x) > \epsilon > 0$ on the interval $(x_0 - \delta, x_0 + \delta)$. Then considering the upper and lower Reimann sums for a partition Q containing the interval $[x_0 - \frac{\delta}{2}, x_0 + \frac{\delta}{2}]$ of width δ, since $f(x) > 0$ on this interval and also $f(x) \geq 0$ everywhere, we have $L(f, Q) \geq \epsilon\delta > 0$. This implies that the integral is greater than zero as

$$0 < L(f, Q) \leq \sup_P L(f; P) = \int_a^b f(x)\,dx$$

which is a contradiction.

To show that we cannot drop the continuity assumption, consider the function

$$g(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

defined on $[0, 1]$ (this is similar to example 2.12 in the notes). Now consider the partitions $P_n = (0, \frac{1}{2^n}, 1)$. We have the upper Riemann sums

$$U(g; P_n) = \frac{1}{2^n} \times 1 + \left(1 - \frac{1}{2^n}\right) \times 0 = \frac{1}{2^n}$$

and lower Riemann sums

$$L(g; P_n) = \frac{1}{2^n} \times 0 + \left(1 - \frac{1}{2^n}\right) \times 0 = 0.$$

Hence the upper Riemann integral has

$$\inf_P U(g; P) \leq \inf_n U(g; P_n) = \inf_n \frac{1}{2^n} = 0$$

and the lower Riemann integral has

$$\sup_P L(g; P) \geq \sup_n L(g; P_n) = 0$$

and therefore

$$\int_0^1 g(x)\, dx = 0$$

whilst g is not continuous.

4. (a) Prove that the function $f : [0, 1] \to \mathbb{R}$ where

$$f(x) = \sqrt{x}$$

is Riemann integrable on $[0, 1]$.

Take the partitions P_n given by the uniformly spaced points $x_i = \frac{i}{n}$ for $i = 0, \ldots, n$.
Then since f is increasing, for the lower Riemann sum we have

$$L(f, P_n) = \sum_{i=1}^{n} \sqrt{\frac{i-1}{n}} \frac{1}{n}$$

and for the upper sum

$$U(f, P_n) = \sum_{i=1}^{n} \sqrt{\frac{i}{n}} \frac{1}{n}.$$

Hence

$$\begin{aligned} U(f, P_n) - L(f, P_n) &= \left| \sum_{i=1}^{n} \sqrt{\frac{i}{n}} \frac{1}{n} - \sqrt{\frac{i-1}{n}} \frac{1}{n} \right| \\ &= \frac{1}{n} \sum_{i=1}^{n} \sqrt{\frac{i}{n}} - \sqrt{\frac{i-1}{n}} \\ &= \frac{1}{n} (\sqrt{1} - \sqrt{0}) \\ &= \frac{1}{n} \end{aligned}$$

where we used the fact that we have a telescopic sum in the 3rd line.
Then for $\epsilon > 0$ we can take all $n \geq N$ with $N > \frac{1}{\epsilon}$, and we have that $U(f, P_n) - L(f, P_n) < \epsilon$, hence Riemann's criterion for integrability is satisfied.

(b) Find the derivative of the function $f : [0, \infty) \to \mathbb{R}$ given by

$$f(x) = \cos\left(\int_0^{\sin(x)} e^{-t^2} \, dt \right)$$

justifying your steps.

Denote

$$F(x) = \int_0^x e^{-t^2} \, dt$$

then since the integrand is continuous, by the FTC we have

$$F'(x) = e^{-x^2}.$$

Now since

$$f(x) = \cos(F(\sin(x)))$$

is a composition of differentiable functions, we can apply the chain rule to obtain

$$\begin{aligned} f'(x) &= -\sin(F(\sin(x)))F'(\sin(x))\cos(x) \\ &= -\sin\left(\int_0^{\sin(x)} e^{-t^2}\,dt\right) e^{-\sin^2(x)}\cos(x). \end{aligned}$$

(c) Prove, using an "ϵ-δ" argument, that the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \dfrac{2x_1x_2}{\sqrt{x_1^2+x_2^2}}, & x \neq (0,0)^\intercal \\ 0, & x = (0,0)^\intercal \end{cases}$$

for all $x = (x_1, x_2)^\intercal \in \mathbb{R}^2$ is continuous at $x = (0,0)^\intercal$. *Hint: you may like to make use of Young's inequality* $|ab| \leq \frac{1}{2}(a^2+b^2)$.

We have that for all $\epsilon > 0$,

$$\left|\frac{2x_1x_2}{\sqrt{x_1^2+x_2^2}} - 0\right| \leq \frac{2\frac{1}{2}\,(x_1^2+x_2^2)}{\sqrt{x_1^2+x_2^2}} = \sqrt{x_1^2+x_2^2} < \delta = \epsilon$$

for all $\|(x_1,x_2)^\intercal\| = \sqrt{x_1^2+x_2^2} < \delta$ with $\delta = \epsilon$.